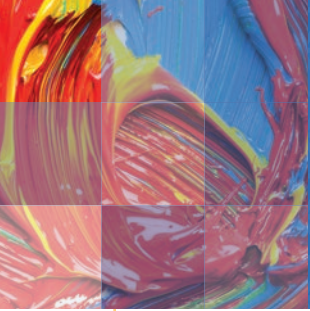




Systems of Linear Equations

When equations with the same variables are grouped together, a system of equations is formed.

Determining Whether a Given Ordered Pair Is a Solution to a System

Definition

A **system of equations** is two or more equations that contain the same variables.

A given ordered pair is a solution to a system if it satisfies all of the equations.

EXAMPLE 1

Determine whether each point is a solution to the system.

$$\begin{cases} 2x - 4y = 8 \\ y = 3x - 7 \end{cases}$$

A $(2, -1)$

SOLUTION

Substitute the coordinates of each point into each equation.

$$\begin{array}{rclcl} 2x - 4y & = & 8 & & y = 3x - 7 \\ 2 \cdot 2 - 4 \cdot (-1) & \stackrel{?}{=} & 8 & & -1 \stackrel{?}{=} 3 \cdot 2 - 7 \\ 4 + 4 & \stackrel{?}{=} & 8 & & -1 \stackrel{?}{=} 6 - 7 \\ 8 & = & 8 & \checkmark & -1 = -1 \checkmark \end{array}$$

The point $(2, -1)$ is a solution to the system because it makes both equations true.

B $(8, 2)$

SOLUTION

Substitute the coordinates of each point into each equation.

$$\begin{array}{rclcl} 2x - 4y & = & 8 & & y = 3x - 7 \\ 2 \cdot 8 - 4 \cdot 2 & \stackrel{?}{=} & 8 & & 2 \stackrel{?}{=} 3 \cdot 8 - 7 \\ 16 - 8 & \stackrel{?}{=} & 8 & & 2 \stackrel{?}{=} 24 - 7 \\ 8 & = & 8 & \checkmark & 2 \neq 17 \end{array}$$

The point $(8, 2)$ is not a solution to the system because it does not make both equations true. ■

Solving Systems by Graphing

One way to solve a system of equations is to graph each equation to determine where they intersect. The point of intersection represents the solution to the system.

EXAMPLE 2

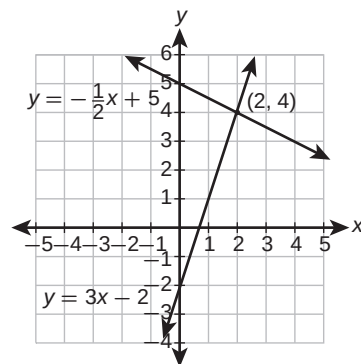
Solve the system by graphing.

A
$$\begin{cases} y = 3x - 2 \\ y = -\frac{1}{2}x + 5 \end{cases}$$

SOLUTION

Graph the two lines. The lines appear to intersect at $(2, 4)$.
You can check that $(2, 4)$ is a solution to both equations.

$$\begin{array}{ll} y = 3x - 2 & y = -\frac{1}{2}x + 5 \\ 4 \stackrel{?}{=} 3 \cdot 2 - 2 & 4 \stackrel{?}{=} -\frac{1}{2} \cdot 2 + 5 \\ 4 \stackrel{?}{=} 6 - 2 & 4 \stackrel{?}{=} -1 + 5 \\ 4 = 4 \checkmark & 4 = 4 \checkmark \end{array}$$



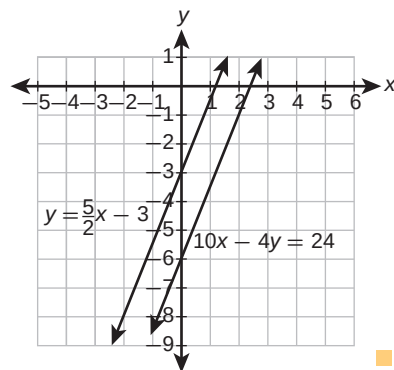
B
$$\begin{cases} 10x - 4y = 24 \\ y = \frac{5}{2}x - 3 \end{cases}$$

SOLUTION

Find the x - and y -intercepts of the first equation.

$$\begin{array}{ll} 10x - 4y = 24 & 10x - 4y = 24 \\ 10 \cdot 0 - 4y = 24 & 10x - 4 \cdot 0 = 24 \\ -4y = 24 & 10x = 24 \\ y = -6 & x = 2.4 \end{array}$$

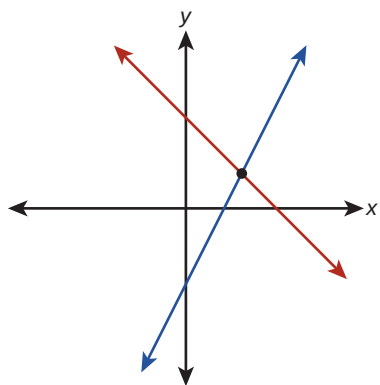
Then graph each line. Because the lines are parallel, they never intersect. This system has no solution.



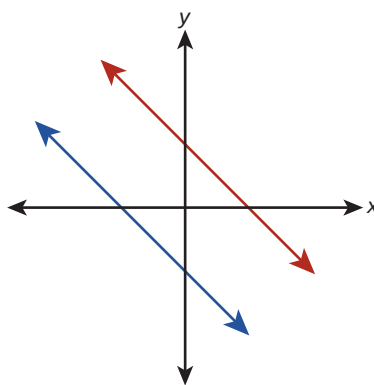
Classifying Systems of Linear Equations

A system of two linear equations can have one solution, no solution, or infinitely many solutions. The table shows how linear systems are classified according to the number of solutions.

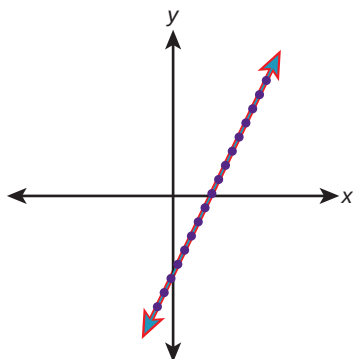
Classifying Systems of Linear Equations



A linear system is **consistent and independent** if its graphs intersect at one point (one solution).



A linear system is **inconsistent** if the lines are parallel (no solution).



A linear system is **consistent and dependent** if the lines coincide (infinitely many solutions).

Solving Linear Systems by the Substitution Method

Algebraic methods, such as the substitution method, can be used to solve linear systems and often yield more accurate results than graphing.

How to Solve a System by the Substitution Method

Step 1 Solve one equation for one of the variables.

Step 2 Substitute the expression obtained from Step 1 in place of the corresponding variable in the other equation, and solve the resulting equation.

Step 3 Substitute the value of the variable from Step 2 into either of the original equations. Solve this equation to find the value of the other variable.

► **Think About It** You can always check your work by making sure that your solution satisfies both equations in the system.

EXAMPLE 3

Solve the system by substitution.

$$\begin{cases} -x + 4y = 6 \\ 3x - 12y = -18 \end{cases}$$

SOLUTION

Step 1 Both equations are in standard form, so solve one of the equations for x or y . It is easiest to solve the first equation for x .

$$\begin{aligned} -x + 4y &= 6 \\ -x &= -4y + 6 \\ x &= 4y - 6 \end{aligned}$$

Step 2 Substitute the expression $4y - 6$ for x in the second equation.

$$\begin{aligned}3x - 12y &= -18 \\3(4y - 6) - 12y &= -18 \\12y - 18 - 12y &= -18 \\-18 &= -18\end{aligned}$$

The equation $-18 = -18$ is true for all real numbers. This means that there are infinitely many solutions.

Step 3 Because there is no single value of y that is a solution, you don't have to complete Step 3.

So the system is consistent and dependent, and there are infinitely many solutions. ■

► **Think About It** When a system solved algebraically results in an equation that is true for all real numbers, the system has infinitely many solutions. When it results in an equation that is not true for any real number, the system has no solution.

Solving Linear Systems by the Linear Combination Method

Another algebraic method to solve a system is the linear combination method.

How to Solve a System by the Linear Combination Method

Step 1 Convert both equations to standard form.

Step 2 Multiply one or both of the equations by a real number so that the coefficient of one variable in one equation is the additive inverse of the coefficient of the same variable in the other equation.

Step 3 Add the equations, and then solve the resulting equation.

Step 4 Substitute the value of the variable from Step 3 into either of the original equations. Solve this equation for the second variable.

► **Remember** The sum of a number and its additive inverse is 0.

► **Think About It** The solution to a linear system of equations does not change when you multiply one or both of the equations by a nonzero real number.

EXAMPLE 4

Solve the system by linear combination.

$$\begin{cases} 2x + 5y = 1 \\ y = \frac{3}{4}x - \frac{13}{4} \end{cases}$$

SOLUTION

Step 1 Convert the second equation to standard form by multiplying through by 4 and subtracting $3x$ from both sides.

$$\begin{cases} 2x + 5y = 1 \\ -3x + 4y = -13 \end{cases}$$

Step 2 Multiply the first equation by 3 and the second equation by 2 so that the coefficients of x are additive inverses.

Step 3 Add the equations to eliminate x . Solve the result for y .

$$\begin{array}{r} 6x + 15y = 3 \\ -6x + 8y = -26 \\ \hline 23y = -23 \Rightarrow y = -1 \end{array}$$

Step 4 Substitute -1 for y in either of the original equations to find x .

$$2x + 5 \cdot (-1) = 1 \Rightarrow 2x - 5 = 1 \Rightarrow 2x = 6 \Rightarrow x = 3$$

So the solution to the system is $(3, -1)$. ■

Solving a System of Three Equations with Three Variables

It is also possible to solve a system of three equations containing three variables. The solution of a system of three equations in three variables can have one solution, infinitely many solutions, or no solution. A system of three equations is solved by finding ordered triples that satisfy all of the equations.

How to Solve a System of Three Equations with Three Variables

Step 1 Convert each equation to standard form.

Step 2 Eliminate one variable from two of the equations.

Step 3 Using these two equations and substitution or linear combination, solve for one of the remaining variables.

Step 4 Substitute the value of the variable from Step 3 into either of the equations in Step 2. Solve this equation for the second variable.

Step 5 Substitute the value of the variables from Step 3 and Step 4 into any of the original equations.

► Think About It

NOTATION The solution of a system of three equations in three variables x, y, z is called an **ordered triple** and is written (x, y, z) .

EXAMPLE 5

Solve the system.

$$\begin{cases} 2x + 5y - z = -8 \\ -x + y + z = -3 \\ -3x - y + 2z = -1 \end{cases}$$

SOLUTION

Step 1 The system has three equations in standard form. Eliminate the z variable from the first two equations by combining them.

$$\begin{array}{rcl} 2x + 5y - z & = & -8 \\ -x + y + z & = & -3 \\ \hline x + 6y & = & -11 \end{array}$$

Step 2 Multiply the second equation by -2 . Then combine the second equation with the third equation to eliminate the z variable.

$$\begin{array}{rcl} 2x - 2y - 2z & = & 6 \\ -3x - y + 2z & = & -1 \\ \hline -x - 3y & = & 5 \end{array}$$

Step 3 Now you have a system of two linear equations with two variables. Combine the equations to solve the system.

$$\begin{array}{rcl} x + 6y & = & -11 \\ -x - 3y & = & 5 \\ \hline 3y & = & -6 \\ y & = & -2 \end{array}$$

Step 4 Use the value of y to solve for x in any of the two equations. Then substitute the value of x and y into one of the original equations to solve for z .

$$\text{Solve for } x: \quad x + 6 \cdot (-2) = -11 \Rightarrow x - 12 = -11 \Rightarrow x = 1$$

$$\begin{aligned} \text{Solve for } z: \quad 2 \cdot 1 + 5 \cdot (-2) - z &= -8 \Rightarrow 2 - 10 - z = -8 \\ &\Rightarrow -8 - z = -8 \Rightarrow z = 0 \end{aligned}$$

The solution to the system is $(1, -2, 0)$. ■

► **Think About It** Check that your solution satisfies every equation in the system.